



EXPERIMENT-5

(a) Deploy a static frontend, integrate it with the API, and add a simple login using Cognito

AIM:

PROCEDURE:

Step-1: Create a frontend folder containing index.html, login.html, style.css, and app.js files.

Step-2: Create the index.html page with a To-Do List interface containing a task input field, Add Task button, task list, and Logout button.

Step-3: Configure app.js with the API Gateway URL to send POST requests for creating tasks and GET requests for retrieving tasks from the serverless API.

Step-4: Open Amazon Cognito, create a User Pool with Email as the sign-in option, create a test user, and note the User Pool ID, App Client ID, and Cognito domain.

Step-5: Create login.html with a Login with AWS Cognito button that redirects the user to the Cognito managed login page using the configured client ID, domain, and callback URL.

Step-6: Configure the Cognito application with `http://localhost:8080/index.html` as the callback URL and `http://localhost:8080/login.html` as the logout URL.

Step-7: Start the frontend using a local HTTP server, access the login page, authenticate through AWS Cognito, and verify that the user is redirected to the To-Do List page.

Step-8: Add a task through the frontend and verify that the task is sent through API Gateway and Lambda and retrieved from the database and displayed in the To-Do List.

PROGRAM:

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>To-Do Application</title>
```



```
<link rel="stylesheet" href="style.css">
</head>

<body>

<h1>To-Do List</h1>

<input type="text" id="task" placeholder="Enter task">
<button onclick="addTask()">Add Task</button>

<ul id="taskList"></ul>

<button onclick="logout()">Logout</button>

<script src="app.js"></script>

</body>
</html>
```

app.js

```
const API_URL = "YOUR_API_URL";

async function addTask() {

    const task = document.getElementById("task").value;

    const response = await fetch(API_URL, {
        method: "POST",

        headers: {
            "Content-Type": "application/json"
        },

        body: JSON.stringify({
            task: task
        })
    });

    const data = await response.json();

    alert("Task Added");

    loadTasks();
}
```



}

```
async function loadTasks() {  
  
    const response = await fetch(API_URL);  
  
    const data = await response.json();  
  
    const list = document.getElementById("taskList");  
  
    list.innerHTML = "";  
  
    data.forEach(item => {  
  
        const li = document.createElement("li");  
  
        li.innerHTML = item.task;  
  
        list.appendChild(li);  
    });  
}
```

```
function logout() {  
  
    localStorage.removeItem("login");  
  
    window.location.href = "login.html";  
}
```

loadTasks();

login.html

```
<!DOCTYPE html>  
<html>  
  
<head>  
    <title>Login</title>  
</head>  
  
<body>  
  
<h2>Login</h2>
```



```
<button onclick="cognitoLogin()">
  Login with AWS Cognito
</button>
```

```
<script>
```

```
function cognitoLogin() {

  const domain =
    "https://ap-south-11rayxgfj3.auth.ap-south-1.amazoncognito.com";

  const clientId =
    "5bqogeq3i3m01drodl18ipuhav";

  const redirectUri =
    "http://localhost:8080/index.html";

  const url =
    domain +
    "/oauth2/authorize" +
    "?client_id=" + clientId +
    "&response_type=token" +
    "&scope=email+openid" +
    "&redirect_uri=" + redirectUri;

  window.location.href = url;
}
```

```
</script>
```

```
</body>
```

```
</html>
```

OUTPUT:

Login

Login with Cognito



Sign in

Sign in to your account.

Email address

Next

New user? [Create an account](#)

Sign up

Create a new account.

Email address

Password

- ✔ Password must be at least 8 characters
- ✔ Use a number
- ✔ Use a lowercase letter
- ✔ Use an uppercase letter
- ⊖ Use a special character

Confirm password

☐ Show password

Sign up

Have an account already? [Sign in](#)



Confirm your account



We have sent a code in an Email message to m***@g***. To confirm your account, enter your code.

Code

[Resend code](#)

[Confirm account](#)

To-Do List

[Add Task](#)

[Logout](#)

To-Do List

[Add Task](#)

- Learn DynamoDB
- Complete CAD lab
- Complete CAD lab

[Logout](#)



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Page No:

RESULT:

NAME: T AGNES HELEN

REG. NO. : 311124205002



EXPERIMENT-5

(b) Deploy a static frontend, integrate it with the API, and add a simple login using Azure AD B2C.

AIM:

PROCEDURE:

Step-1: Login to the Azure Portal, search for Azure AD B2C, and create an Azure AD B2C tenant.

Step-2: Open the created B2C tenant, navigate to App registrations, and register a new application named Todo Frontend.

Step-3: Select Accounts in any identity provider, add <http://localhost:5500/index.html> as the redirect URI, and register the application.

Step-4: Note the generated Application (Client) ID and Tenant Name for configuring the login page.

Step-5: Navigate to Azure AD B2C → User Flows → New User Flow, select Sign up and sign in, configure Email and Password, and create the user flow.

Step-6: Add a Login with Microsoft Azure button to login.html and configure the tenant name, client ID, user flow, and redirect URI.

Step-7: Configure the Azure login URL using the OAuth 2.0 authorization endpoint and redirect the user to Microsoft Azure for authentication.

Step-8: Run the frontend using the local HTTP server, access the login page, select Login with Microsoft Azure, and verify that successful authentication redirects the user to the To-Do List page.

PROGRAM:

login.html

```
<!DOCTYPE html>
<html>
```



```
<head>
  <title>Login</title>
</head>

<body>

<h2>Login</h2>

<button onclick="azureLogin()">
  Login with Microsoft Azure
</button>

<script>

function azureLogin() {

  const tenant =
    "YOUR_TENANT_NAME";

  const clientId =
    "YOUR_CLIENT_ID";

  const policy =
    "YOUR_USER_FLOW";

  const redirectUri =
    "http://localhost:5500/index.html";

  const loginUrl =
    "https://" +
    tenant +
    ".b2clogin.com/" +
    tenant +
    ".onmicrosoft.com/" +
    policy +
    "/oauth2/v2.0/authorize" +
    "?client_id=" + clientId +
    "&response_type=id_token" +
    "&redirect_uri=" + redirectUri +
    "&scope=openid" +
    "&response_mode=fragment";

  window.location.href = loginUrl;
}
```



</script>

</body>

</html>

OUTPUT:

Login Page



Microsoft Azure AD B2C Login



Successful Authentication



To-Do List Page

RESULT: